

Mrinal Kanti Giri, Ph.D.

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Research Profile

Theoretical physicist working on non-equilibrium dynamics of interacting quantum many-body systems, combining tensor-network numerics (MPS, DMRG, TEBD, TDVP) with digital quantum simulation on superconducting NISQ hardware. My research line runs from interacting quantum walks, flux and localization in ladder geometries (Ph.D.) to executing these models on real quantum processors, with circuit compression and error mitigation (postdoc).

Academic Appointments

Research Fellow, Nanyang Technological University, Singapore May 2025 – present

- Develop digital quantum simulation algorithms for non-equilibrium dynamics and emergent phases in strongly correlated spin systems
- Run and benchmark circuits on IBM Quantum hardware with circuit compression and error mitigation

Postdoctoral Research Fellow, National Tsing Hua University, Taiwan Oct 2023 – Apr 2025

- Led a project on digital quantum simulation of flat-band and all-bands-flat transport (first and corresponding author, *Quantum Sci. Technol.* 2026)
- Developed tensor-network workflows (MPS, DMRG, TEBD, TDVP) as classical benchmarks for hardware runs

Postdoctoral Fellow, CQuERE, TCG-CREST, Kolkata, India Jul 2022 – Sep 2023

- Realized topological quantum walks on a NISQ digital quantum computer (first and corresponding author, *Physica Scripta* 2025)

Education

Ph.D. in Physics, Indian Institute of Technology Guwahati 2018 – 2022

Thesis: *Quantum dynamics of low-dimensional interacting systems using the continuous-time quantum walk*. Advisor: Dr. Tapan Mishra.

M.Sc. in Physics, Indian Institute of Technology Guwahati 2016 – 2018

Project: *Quantum phase transition in a multicomponent Bose–Einstein condensate*

Publications

9 papers (8 peer-reviewed, 1 preprint) · first author on 7 · † corresponding author

Preprints

1. M. K. Giri[†], S. Tai-Ming, P. Sengupta, and P. Chen, “Quantum simulation of detuning-controlled proximity-induced localization and reentrant delocalization in a quasiperiodic ladder,” [arXiv:2609.37059](https://arxiv.org/abs/2609.37059) (2026).

Peer-reviewed journal articles

2. M. K. Giri[†] and P. Chen, “Digital quantum simulation of flat-band and all-bands-flat dynamics for tunable quantum transport,” *Quantum Sci. Technol.* **11**, 035064 (2026).

3. **M. K. Giri**[†] and S. B. Mandal, “Realizing topological quantum walks on a NISQ digital quantum computer,” *Physica Scripta* **100**, 035111 (2025).
4. **M. K. Giri**, B. Paul, and T. Mishra, “Flux enhanced localization and reentrant delocalization in the quantum walk of interacting bosons on a two-leg ladder,” *Phys. Rev. A* **109**, 043308 (2024).
5. **M. K. Giri**, B. Paul, and T. Mishra, “Flux-induced reentrant dynamics in the quantum walk of interacting bosons,” *Phys. Rev. A* **108**, 063319 (2023).
6. A. Padhan, R. Parida, S. Lahiri, **M. K. Giri**, and T. Mishra, “Quantum phases of constrained bosons on a two-leg Bose–Hubbard ladder,” *Phys. Rev. A* **108**, 013316 (2023).
7. **M. K. Giri**, S. Mondal, B. P. Das, and T. Mishra, “Signatures of nontrivial pairing in the quantum walk of two-component bosons,” *Phys. Rev. Lett.* **129**, 050601 (2022).
8. A. Padhan, **M. K. Giri**, S. Mondal, and T. Mishra, “Emergence of multiple localization transitions in a one-dimensional quasiperiodic lattice,” *Phys. Rev. B* **105**, L220201 (2022).
9. **M. K. Giri**, S. Mondal, B. P. Das, and T. Mishra, “Two-component quantum walk in one-dimensional lattice with hopping imbalance,” *Scientific Reports* **11**, 22056 (2021).

Talks and Presentations

Invited talk

- *Emerging Trends and Recent Advances in Quantum Computing and Quantum Technologies*, NMIT, Bengaluru, India · Mar 2026

Contributed talks

- APS March Meeting, Colorado, USA · Mar 2026
- Young Investigators Meet on Quantum Condensed Matter Theory (YIMQCMT), NISER, India · Oct 2022
- Annual Conference on Quantum Condensed Matter (QMAT), IIT Kanpur, India · Sep 2022

Posters

- ICTP–IBM School on Quantum Computing and Simulation, ICTP, Trieste, Italy · May 2026
- Recent Progress in Many-Body Theories (RPMBT22), University of Tsukuba, Japan · Sep 2024
- Ultracold Atoms Japan 2024, OIST, Okinawa, Japan · Apr 2024
- 3rd International Quantum Tensor Network (IQTN) Plenary Meeting, TUM Raitenhaslach, Germany · Jul 2023
- Fundamental Sciences & Quantum Technologies using Atomic Systems, PRL Ahmedabad, India · Sep 2020

Mentoring and Teaching

Student co-mentoring, NTU Singapore and NTHU Taiwan

2023 – present

Co-mentor M.Sc. and Ph.D. students on digital quantum simulation, tensor-network methods, and IBM Quantum hardware with Qiskit; guide students through manuscript preparation.

Teaching Assistant, IIT Guwahati

2018 – 2022

Tutorials and laboratory sessions in Quantum Mechanics, Electromagnetism, and Computational Physics.

Professional Service

- Referee for *Quantum Science and Technology* and *Physica Scripta* (IOP Publishing)

Awards and Fellowships

- CSIR-NET (JRF), All India Rank 90 · 2018
- DST INSPIRE Scholarship, top 1% of students in West Bengal · 2014

Technical Expertise

- **Quantum computing:** Trotterized time evolution, circuit compression and optimization, error mitigation, noise modelling, VQA, QITE, hardware execution on IBM Quantum
- **Numerical methods:** MPS, DMRG, TEBD, TDVP, exact diagonalization, cluster mean-field theory, stochastic series expansion QMC
- **Software:** Python, Qiskit, Qiskit Runtime, Git, Linux, HPC clusters

References

Available on request.